

PERVOMAYSKOYE, Russian Federation

WMO: 293480

Lat: 57.0753N

Lon: 86.2374E

Elev: 375

StdP: 14.50

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.1	-29.1	-40.1	0.6	-34.7	-34.2	0.8	-29.2	16.2	13.9	14.1	14.0	1.5	230	0.572

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	84.6	67.6	81.6	66.3	78.6	64.8	70.4	79.9	68.5	77.8	66.6	75.7	4.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	101.7	75.3	65.1	94.0	73.2	63.1	87.5	71.5	34.7	80.3	33.1	78.1	31.5	75.8	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	13.7	11.7	10.0	DB	-39.5	89.5	6.8	2.7	-44.3	91.5	-48.3	93.1	-52.1	94.6	-57.0	96.6
(5)				WB	-39.8	73.1	6.6	2.3	-44.6	74.7	-48.4	76.1	-52.1	77.4	-56.8	79.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.0	-1.3	5.7	21.2	36.3	50.1	62.5	66.2	60.5	48.7	35.2	16.5	4.3
	DBStd	25.79	16.07	13.85	11.54	9.80	10.01	8.07	5.85	6.71	8.14	9.52	14.53	15.89
	HDD50	7296	1592	1239	892	421	127	9	0	5	123	466	1005	1417
	HDD65	11525	2057	1659	1357	861	472	137	56	172	492	925	1455	1882
	CDD50	1447	0	0	0	9	129	386	504	330	83	6	0	0
	CDD65	202	0	0	0	0	10	64	95	31	2	0	0	0
	CDH74	1921	0	0	0	2	149	634	822	289	25	0	0	0
	CDH80	443	0	0	0	0	36	157	192	55	3	0	0	0
Wind	WSAvg	4.3	4.5	4.7	5.1	5.2	4.7	3.4	2.8	3.0	3.6	4.9	5.1	5.1
Precipitation	PrecAvg	18.6	0.9	0.7	0.7	1.0	1.6	2.3	2.6	2.4	2.0	1.6	1.6	1.3
	PrecMax	25.5	1.8	1.4	2.2	1.9	3.7	4.8	5.0	5.1	4.1	3.0	2.3	2.5
	PrecMin	13.5	0.2	0.1	0.1	0.3	0.4	0.2	0.3	0.6	0.6	0.5	0.7	0.4
	PrecStd	3.0	0.4	0.4	0.5	0.4	0.9	1.1	1.2	0.9	0.9	0.6	0.5	0.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	32.1	34.3	49.0	68.3	84.9	88.5	89.2	85.6	77.5	62.7	44.8	34.0
		MCWB	29.9	31.5	40.3	53.8	64.6	66.8	69.4	69.8	62.9	49.9	41.7	32.1
	2%	DB	28.2	31.7	43.5	62.2	78.1	84.2	84.8	80.9	71.4	57.1	40.0	31.3
		MCWB	26.8	29.3	36.0	49.6	60.2	65.7	69.4	66.4	59.1	48.1	37.3	29.9
	5%	DB	25.3	28.4	39.5	56.8	73.2	81.0	81.7	77.1	66.5	52.7	36.7	28.3
		MCWB	23.6	26.3	33.6	46.4	57.6	64.6	67.8	64.7	56.6	46.4	34.4	26.7
	10%	DB	21.4	24.8	36.4	51.6	67.8	77.4	78.7	73.2	61.9	48.2	33.8	24.5
		MCWB	20.1	22.8	32.1	43.3	54.8	63.4	66.2	62.9	54.0	43.8	32.2	23.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.4	32.4	40.9	54.5	66.5	70.5	74.4	71.9	63.7	52.8	42.1	32.5
		MCDB	31.8	33.6	48.1	66.3	83.2	81.8	82.6	81.5	74.4	58.6	44.3	33.4
	2%	WB	26.9	29.5	36.9	50.6	62.0	68.2	71.6	68.7	60.2	49.5	37.6	29.8
		MCDB	28.1	31.5	41.9	60.4	75.1	79.1	81.0	77.7	69.6	55.4	39.7	30.8
	5%	WB	23.6	26.3	34.6	46.9	58.7	66.4	69.6	66.2	57.5	46.9	34.6	26.7
		MCDB	25.1	28.2	38.6	55.7	70.4	77.5	78.4	74.2	64.4	52.3	36.3	28.1
	10%	WB	20.1	23.0	32.4	43.3	55.7	64.7	67.8	64.1	55.0	43.8	32.3	23.3
		MCDB	21.3	24.7	36.1	50.9	66.0	74.9	76.3	71.5	60.9	48.0	33.7	24.6
Mean Daily Temperature Range	5% DB	MDBR	13.4	15.5	17.5	17.2	19.1	18.6	18.4	17.7	16.1	11.7	11.0	12.3
		MCDBR	11.9	12.6	18.3	24.5	26.8	24.9	23.1	23.9	24.2	19.4	11.2	13.3
		MCWBR	11.5	11.4	13.0	14.9	13.6	10.3	9.8	11.2	12.7	12.0	10.1	12.9
	5% WB	MCDBR	12.2	11.5	16.1	23.3	24.7	21.3	20.1	20.4	21.4	17.6	11.0	13.1
		MCWBR	12.0	10.9	12.5	15.1	14.1	10.2	10.1	10.9	12.2	11.6	10.1	12.8
Clear-Sky Solar Irradiance	taub	0.289	0.322	0.370	0.396	0.406	0.405	0.439	0.406	0.370	0.347	0.314	0.289	
	taud	1.989	1.970	1.913	2.060	2.215	2.309	2.202	2.283	2.331	2.239	2.030	1.940	
	Ebn at Noon	169	216	238	256	263	265	252	252	244	212	156	128	
	Edh at Noon	20	32	44	45	41	38	41	36	29	24	19	16	
All-Sky Solar Radiation	RadAvg	226	535	969	1391	1554	1832	1678	1322	843	420	200	131	
	RadStd	27	44	74	100	154	183	117	99	116	37	33	23	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page